PLEASE NOTE THAT THIS LAB QUIZ IS 2 PAGES AND WITH A TOTAL OF 10 POINTS.

Exam Instructions:

- Open NetBeans IDE.
- From File, close all projects.
- Create a new Java project named **quizProjectSec2**.
- Inside the **quizProjectSec2** project, add a new package named **quiz1**.
- Within the **quiz1** package, create the required classes as specified in the exam question.
- Design a class named **quiz1** that contains the following:

Question 1

[5 POINTS]

A. write a java a method named **calculateTotalPrice** that:

- Asks the user to enter a price of an item (double).
- Asks the user to enter a discount rate (ex: 0.05) (double).
- Uses the following formula to calculate the total price:

  totalPrice = price – (price * discountRate)
- Returns the total price as a double.

B. In the main method:

- Call calculateTotalPrice.
- Print the calculated total price.

Question 2 **[5 POINTS]**

Write a Java Method that verifies the validity of a time based on simple rules.

1. Create a method named **checkTime** that:
   - Accepts a String parameter representing a time (ex: 11:00 AM or 11:00 am).
   - Checks the following rules and returns an **integer code:**
     - **2** → if the time contains ":" and "AM".
     - **1** → if the number contains ":" only.
     - **0** → if the number contains neither ":" and "AM".
2. **In the main method:** String صل دي فقط
   - Prompt the user to enter a time.
   - Pass the entered string to the checkTime method.
   - Display(print) a message based on the returned value:

Returned Value	Message
2	"Valid time: contains : and AM"
1	"Time contains : only — missing AM"
0	"Invalid time — missing : or AM"

Output Sample:

Input: 09:00 am
Output: Valid time: contains : and AM

Input: 09:00
Output: Time contains : only — missing AM

Input: 0900 AM
Output: Invalid time — missing : or AM